

Reminder: Empirical and simulation exercises must be submitted with “stand-alone” code and data (if any) in an appropriate format. The homework is due Sunday (data, code, and word-processed report by email to **the instructor before 10 PM Sunday**). Also, submit your work-processed report via **Gradescope**.

1. Download quarterly real GDP and updated Romer and Romer monetary policy shock series. You can download the updated monetary policy shock series that Johannes Wieland and Mu-Jeung Yang use for the paper titled “Financial Dampening” from Wieland’s webpage. Note that the sample ends in 2007 because of the binding ZLB after then.

- a) Estimate the responses of real GDP to a one standard deviation contractionary monetary policy shock using local projections for horizons $h = 0, \dots, 20$. Construct the 90 percent confidence interval for your impulse responses. Try the following model:

$$y_{t+h} - y_{t-1} = c^{(h)} + \sum_{m=0}^M \beta_m^{(h)} x_{t-m} + \sum_{q=1}^Q \phi_q^{(h)} \Delta y_{t-q} + u_{t+h|t-1},$$

where $M = Q = 4$.

- b) Repeat a) under the minimum delay assumption. That is, assume that the contemporaneous response is zero. For this purpose, you can further include Δy_t on the right side, and estimate the model for $h \geq 1$.
 - c) Repeat the exercise for the TFP series and the utilization-adjusted TFP series, available from John Fernald’s webpage. Note that his Excel file includes $\Delta \log(TFP_t)$. Thus, it should be cumulated to have a (log) TFP series. Does the measured TFP respond endogenously to monetary policy shocks? How about the utilization-adjusted TFP series? Which variable is a more reasonable proxy for productivity shocks?
2. Consider an ARMA(1,1) model $y_t = \alpha y_{t-1} + \epsilon_t + \theta \epsilon_{t-1}$, where $\epsilon \sim iid N(0, \sigma_\epsilon^2)$.
 - a) Write this model in a state-space representation.
 - b) Suppose $\alpha = 0.8, \theta = 0.4, \sigma_\epsilon^2 = 3$. Generate artificial data for sample sizes $T=50, 100$, and 500. Using the simulated data, estimate the model with the Kalman filter.
 - c) Repeat the exercise 1,000 times and compute mean bias, variance and MSE of the parameter estimates. Compare your results with what you obtained using the Hannan-Rissanen procedure and MoM in homework 2.